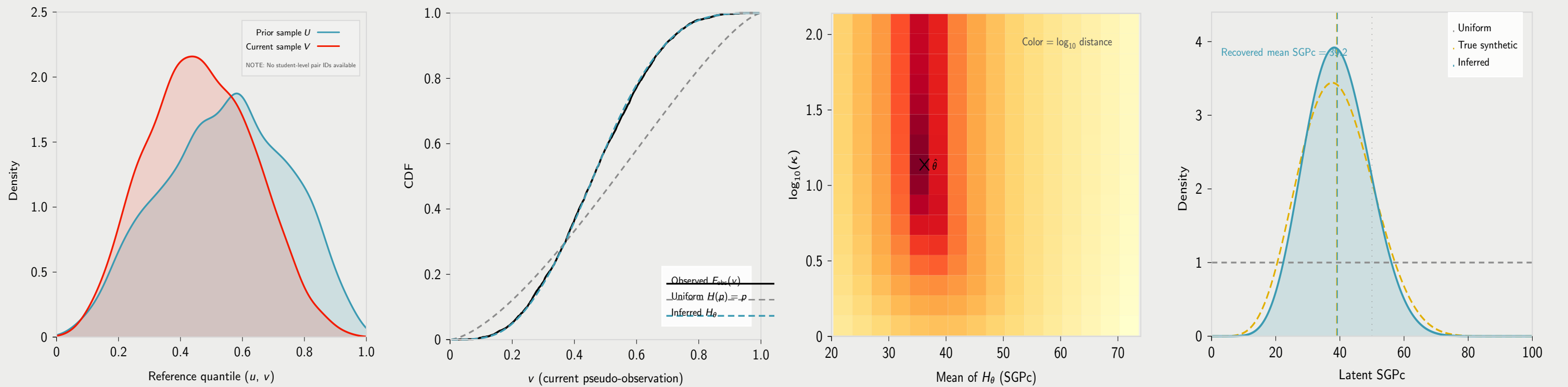


Stochastic Growth: From observed *V* to inferred *SGPcFlow* growth regime

Synthetic low-growth case (true mean SGPc 39), baseline *t*-copula kernel ($\rho = 0.72$, $df = 8$)



A. Independent cross-sectional inputs

The starting point is two independent cross-sectional samples: prior-grade reference percentiles $U = F_X^{\text{ref}}(x_i)$ and current-grade reference percentiles $V = F_Y^{\text{ref}}(y_j)$. Crucially, there is *no student-level linkage* between these samples—we cannot match any u_i to its corresponding v_j . This is exactly the situation faced by international assessments such as TIMSS, where Grade 4 and Grade 8 cohorts are sampled independently. The distributions of U and V are the only observables; all subsequent inference proceeds from these marginals alone.

B. The analytic identity

Given a candidate growth regime H_{θ} and the baseline transition kernel $F_0(v|u)$ from the fitted copula, the **analytic identity**

$$F_{\theta}(v) = \mathbb{E}_U[H_{\theta}(F_0(v|U))] = \frac{1}{n} \sum_{i=1}^n H_{\theta}(F_0(v|u_i))$$

predicts the current-grade CDF *exactly*—no Monte Carlo simulation is needed. The uniform regime $H(p) = p$ (grey dashed) systematically misses the observed CDF, while the inferred $H_{\hat{\theta}}$ (teal dashed) closes the gap.

C. Minimising the distance

We search over candidate parameter vectors $\theta = (\text{mean}, \kappa)$ of the Beta growth regime, computing the Wasserstein-1 distance

$$W_1(F_{\text{obs}}, F_{\theta})$$

at each grid point. The heatmap shows $\log_{10}$ distance: dark regions are far from the observed CDF; light regions are close. A two-stage procedure—coarse grid scan followed by L-BFGS-B refinement—locates $\hat{\theta}$, the parameter vector that minimises the distance between the predicted and observed marginal CDFs of V .

D. The recovered growth regime

The estimated growth regime $H_{\hat{\theta}}$ (teal) is a Beta density on $[0, 1]$ representing the distribution of latent conditional growth percentiles (SGPc). It closely matches the true synthetic regime (amber dashed), recovering both the mean and spread. The regime separates the subgroup's *growth occupancy*—where students actually fall in the conditional growth distribution—from the baseline *dependence template* encoded by the copula kernel $F_0(v|u)$. A regime centred below 50 indicates growth that is systematically lower than the reference population.

Flow A → B → C → D | $W_1(F_{\text{obs}}, F_{\theta})$: uniform = 0.0915, inferred = 0.003 (96.7% reduction)